for binary black hole merging, 359f
for travel to another universe, 456, 457f
for quantum foam, 487f, 495f
for a wormhole, 484–85, 485f, 501f
for creation of a wormhole, 496f
Entropy:
definition of, 423, 424b
increase of, 422–26, 424b
of black hole, 423–26, 442, 445–46
Equation of state:
concept of, 153, 193b
of white-dwarf matter, 150–53, 154f, 156b–57b, 200b–201b
of nuclear matter, 193b–95b, 200b–202b
of cold, dead matter, 200b–202b
Equivalence principle, 44–45, 97–98, 99f, 100, 103b, 109, 371
ETH (Zurich Politechnikum), 60, 62, 68, 69, 71, 93, 113, 115
Exotic material:
nature of, 488–90, 508
vacuum fluctuations as, 491–92, 498, 583N491, 583N492
required to hold a wormhole open, 488, 489b, 490–91, 493–94, 498, 504
required when creating a time machine, 499n
Experimental physicists contrasted with other scientists, 319–21, 328

Fifth force, 64
Fission, nuclear:
concept of, 221b–22b
discovery of, 220
Bohr–Wheeler theory of, 220–23
see also Chain reactions
Free will, 509
Fusion, nuclear:
concept of, 183b, 221b–22b
keeps stars hot, 182, 184, 191
as possible power source for quasars, 340
cold fusion, 64

Gamma rays:
as part of electromagnetic spectrum, 25f
emitted by gas accreting into black hole, 26
emitted in black-hole evaporation, 447–48
General covariance, 114n
General relativity:
Einstein’s struggle to formulate, 96–117
domain of validity, 84–85
experimental tests of, 57–58, 116–17, 393, 563N57–58, 568N131
marriage with quantum mechanics, *see* Quantum gravity
see also specific concepts: Einstein field equation; Equivalence principle; Gravitational time dilation; Gravity, tidal; Relativity, principle of; Space warpage; Spacetime curvature; Time warpage
Geodesic, 108–12, 108f, 118b, 126–27, 401
Global methods, 465, 490, 490n, 491